

# Keerthi Rapolu

+1 (281) 802-0700 | [keerthiofrapolu@gmail.com](mailto:keerthiofrapolu@gmail.com) | Houston, TX, USA | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

## Professional Summary

With over 13 years of experience, I've worked across the full evolution of data platforms—from mainframe systems and Oracle databases to modern cloud-based architectures. My background spans Snowflake, Databricks, PL/SQL, and mainframe technologies, with a strong focus on building production-ready data pipelines, designing scalable data warehouses, and leading complex migrations from legacy systems to the cloud. I've delivered large-scale data migrations from Oracle and SQL Server to Snowflake, modernized mainframe workloads using PySpark and Databricks, and implemented automated, reliable workflows using tools like dbt, Airflow, and AWS services.

In recent years, my work has expanded into AI-driven data engineering, supporting analytics, AI, and LLM-based use cases with high-quality, well-governed data. I've contributed to entity resolution using AI/ML techniques, built data models optimized for advanced analytics, and ensured data readiness for downstream consumption by BI tools, AI platforms, and large language models. I bring a practical, end-to-end approach—combining deep legacy expertise with modern cloud, AI, and LLM-focused data solutions.

## Education

**Jawaharlal Nehru Technological University**  
*Bachelor's, Engineering*

**May 2007 - May 2011**

## Certifications

[SnowPro Core](#)

[AWS Certified Solutions Architect – Associate](#)

## Professional Experience

### ServiceNow

*Staff Data Engineer (Cloud FinOps / AI)*

**Houston, TX**

*Feb 2026 – Present*

- Leading the design and development of AI-driven FinOps analytics platforms to deliver cost visibility, optimization, and governance across multi-cloud environments (AWS, Azure, GCP).
- Architecting scalable cost allocation frameworks enabling accurate attribution, financial transparency, and improved cloud spend accountability across business units.
- Defining and evolving data models, platform standards, and processing frameworks using SQL, Python, dbt, Trino, and Apache Iceberg to support high-volume, distributed analytics.
- Building and operating resilient, production-grade data pipelines and orchestration systems using Argo Workflows, ensuring reliability, scalability, and performance.
- Driving adoption of modern data engineering practices, including AI-assisted (LLM-driven) development, to accelerate delivery and improve analytical accuracy.
- Designing multi-cloud billing ingestion and normalization frameworks to standardize cost and usage data across AWS, Azure, and GCP, enabling unified reporting and analysis.
- Developing FinOps data models to support unit economics, cost per resource, and cost attribution across services, applications, and teams.
- Collaborating cross-functionally with engineering, finance, and platform teams to influence FinOps strategy, improve cost efficiency, and deliver actionable insights for decision-making.
- Enabling advanced analytics use cases, including anomaly detection, cost forecasting, and natural language-driven insights on cloud spend data.

### Genpact Ltd.

*Lead Consultant - Client: Kaiser Permanente*

**Houston, TX**

*May 2025 – Jan 2026*

- Designed, built, and maintained scalable data engineering solutions supporting analytics, AI, and customer experience insights across healthcare use cases.
- Delivered an end-to-end solution showcasing Amazon Connect capabilities, including intelligent call routing, sentiment detection, and integration with patient data, receiving strong stakeholder feedback.
- Designed and implemented AWS Glue ETL pipelines to ingest CSV files from Amazon S3 into Amazon RDS (Oracle) and Snowflake, automating data ingestion and modeling into a star schema.
- Developed end-to-end ETL pipelines using AWS Glue (PySpark, Python) to ingest, transform, and model structured data from Amazon S3 into Amazon RDS (Oracle) and Snowflake, implementing fact and dimension tables.
- Built an Amazon Connect Agent Workspace “Patient 360” dashboard displaying schedules, lab results, triage, and billing data using synthetic datasets to ensure HIPAA compliance.

- Designed and managed Snowflake objects including databases, schemas, tables, views, stages, file formats, and tasks; implemented role-based access control (RBAC), optimized warehouse utilization, and enabled scalable analytics and AI-ready workloads.
- Developed AWS Lambda APIs with VPC integration to retrieve patient data from Amazon RDS and Snowflake; integrated APIs into Amazon Connect flows and widgets with CloudWatch monitoring.
- Documented use cases highlighting Amazon Connect AI capabilities; generated synthetic datasets and supporting data models for analytics such as sentiment analysis, call evaluation, and customer experience scoring.
- Engineered robust data pipelines using AWS Glue, Python, and SQL to automate ingestion of CSV files from Amazon S3 into Amazon RDS and Snowflake, improving data accessibility.
- Leveraged Snowflake Cortex to enable advanced analytics and AI use cases, including natural language processing and sentiment analysis on interaction data.
- Integrated Amazon Connect, Amazon Lex, and AWS Bedrock with backend data platforms to support intelligent call routing, sentiment detection, and AI-driven insights.
- Designed data models and curated datasets to support KPIs related to call performance, sentiment, patient engagement, and operational efficiency.

## Landrys

*Lead Data Engineer*

**Houston, TX**

*August 2024 - May 2025*

- Worked in an environment leveraging Databricks, Snowflake, Python, SQL, PySpark, Azure DevOps, Fivetran, Splink, Azure, Azure Data Factory (ADF), SQL Server, and SSIS.
- Developed and optimized Delta Live Tables in Databricks for real-time and streaming data processing; built custom Python and PySpark transformations to clean and prepare datasets for analytics.
- Wrote and executed complex SQL queries to support accurate reporting, analytics, and data validation.
- Automated CI/CD pipeline deployments using Azure DevOps, improving workflow efficiency and reducing manual intervention.
- Integrated multiple data sources into Azure using Fivetran; connected Snowflake with Azure pipelines to enable automated ingestion and near real-time data processing.
- Designed scalable ETL pipelines using Snowflake and Azure Data Factory to handle ingestion, transformation, and loading into data lakes and warehouses.
- Implemented Splink-based entity resolution using machine learning techniques to match and deduplicate records; customized matching logic to reduce false positives and improve precision of business-critical datasets.
- Integrated Splink with Azure pipelines to enable continuous and automated data cleansing workflows.
- Designed and implemented ETL solutions using SSIS integrated with SQL Server to support data migration, synchronization, and hybrid (on-prem and cloud) data integration.

## Impetus Technologies

*Module Lead Engineer - Client: Ally*

**Hyderabad, TG, India**

*December 2022 - August 2024*

- Worked in an environment leveraging Snowflake, Oracle, AWS, GitLab, Qlik Replicate, dbt, Tosca, DataStage, SQL, and Python.
- Developed migration strategies to transition large-scale on-premise Oracle databases to Snowflake; assessed schemas, dependencies, and risks to define mitigation plans.
- Led end-to-end data migration using Qlik Replicate, ensuring minimal downtime and business disruption.
- Engineered and deployed Snowflake objects including tables, views, sequences, synonyms, stored procedures, functions, tasks, streams, stages, and data-sharing configurations aligned with enterprise architecture.
- Designed warehouse configurations and optimized Snowflake environments to support migrated Oracle workloads.
- Leveraged dbt to automate transformations, implement incremental models, enforce version control, and maintain documentation.
- Migrated complex PL/SQL logic into Snowflake using dbt and Snowpark, improving performance and maintainability.
- Utilized Snowpark and optimized SQL queries to process large-scale datasets efficiently.
- Implemented automated testing using Tosca to validate data integrity, enforce business rules, and ensure regulatory compliance.

## Impetus Technologies

*Module Lead Engineer - Client: Norfolk Southern*

**Hyderabad, TG, India**

*January 2022 - December 2022*

- Worked in an environment spanning Databricks, PySpark, JCL, COBOL, Python, Kafka, SQL, AWS, DB2, and Azure.
- Analyzed legacy systems (JCL, COBOL, DB2) to identify data and logic required for migration to Databricks.
- Designed and built Delta Live Tables pipelines for real-time data processing and streaming analytics using Delta Lake.
- Refactored legacy COBOL logic into Python and PySpark, improving performance and maintainability.
- Collaborated with architects to define lakehouse architecture (raw and curated layers) and built transformation pipelines for standardized datasets.
- Integrated Kafka with Databricks to enable real-time ingestion and streaming insights.
- Developed unit tests and validation frameworks to ensure data accuracy, reliability, and code quality.

## Capricorn Software

*Senior Software Engineer - Client: Meridian Cooperative*

**Hyderabad, TG, India**

*February 2019 - January 2022*

- Worked in an environment including Snowflake, Oracle, AWS, SQL, Apache NiFi, Python, AWS Glue, and Lambda.
- Designed and implemented Oracle-to-Snowflake migration pipelines using Snowpipe for near real-time ingestion.
- Utilized Apache NiFi to extract on-premise data, generate files, and load into Amazon S3.
- Implemented Airflow for orchestration, scheduling, and monitoring of migration workflows.

- Created and managed Snowflake objects including databases, schemas, tables, views, sequences, stored procedures, file formats, storage integrations, stages, and Snowpipes.
- Developed AWS Lambda functions to trigger Glue jobs based on S3 events; built ETL pipelines in Glue Studio to load data into S3, Redshift, and RDS.
- Collaborated with cross-functional teams to gather requirements and ensure alignment with business needs.
- Optimized complex SQL queries and conducted end-to-end testing to validate data accuracy and pipeline performance.

**Capricorn Software**  
*Software Engineer - Client: SEDC*

**Hyderabad, TG, India**  
*December 2016 - January 2019*

- Worked across Oracle, MF COBOL, .NET, and Linux environments to deliver enterprise applications.
- Translated business requirements into detailed technical designs, improving communication between stakeholders and development teams.
- Developed features, enhancements, and bug fixes across multiple technologies.
- Created Unix shell scripts to automate operational tasks and improve efficiency.
- Performed unit testing to ensure functionality, reliability, and code stability.

**Blueliner**  
*Software Engineer - Client: MSC*

**Hyderabad, TG, India**  
*January 2012 - November 2016*

- Worked in a mainframe environment using COBOL, JCL, DB2, VSAM, SPUFI, TSO/ISPF, SQL, CICS, ChangeMan, Xpeditor, and File-AID.
- Developed and tested COBOL programs integrated with DB2 databases, supporting critical business operations.
- Followed coding standards and best practices to maintain a reliable and maintainable codebase.
- Collaborated with team members to troubleshoot issues and resolve technical challenges.
- Conducted thorough unit testing to identify defects and ensure overall system stability.

## Skills

**Skills:** Snowflake, Databricks, Oracle PL/SQL, Mainframe Technologies, Data Migration, AWS S3, CSV, Parquet, JSON, dbt (Data Build Tool), PySpark, Qlik Replicate, SQL, Tosca, Python, AWS Redshift, AWS CloudWatch, AWS Glue, AWS Lambda, Apache Airflow, Apache NiFi, IBM InfoSphere DataStage, Informatica, Apache Kafka, Fivetran, Splink, Linux/Unix, Microsoft Azure, DB2, Postman, Data visualization, Tableau, Power BI, COBOL, JCL, VSAM, CICS, Azure DevOps, SQL Server, SQL Server Integration Services (SSIS), Data modeling, ETL, Data integration, Machine Learning, Entity resolution, AWS, LLM, RAG, ChromaDB